BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 07-02-2026**

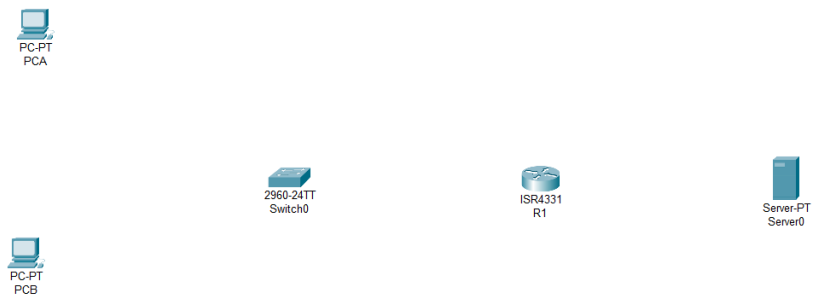
Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 06

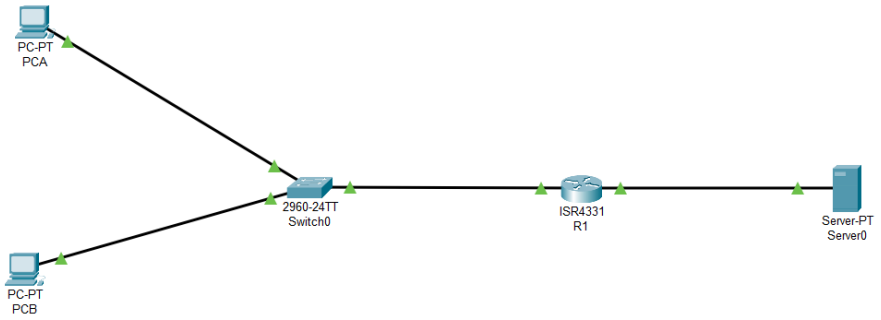
Q. Configure IP ACLs and IPV6 ACLs to Mitigate Attacks.

Steps:

1. Open the Cisco packet
2. Add the two PCs(PCA and PCB), one switch of 2960-24TT(Switch0), one router of 1941(R1), and one server(Server0)



3. Now connect copper straight-through wire to devices as given below.
   - **PCA → Switch0:** PCA – FastEthernet0 → Switch0 FastEthernet0/1
   - **PCB → Switch0:** PCB – FastEthernet0 → Switch0 FastEthernet0/2
   - **Switch0 → Router(R1):** Switch0 FastEthernet0/3 → R1 GigabitEthernet0/0/0
   - **Router(R1) → Server0:** R1 GigabitEthernet0/1 → Server0 FastEthernet0



4. Assign the addresses as follow:
   - To assign addresses to Server0, PCA and PCB go to the desktop → IP configuration.
   - For R1 go to config.

IP Config	IPv4 Address	Subnet Mask	Default Gateway	IPv6 Address	Default Gateway
PCA	192.168.0.2	255.255.255.0	192.168.0.1	2001:DB8:1:10::9	FE80::1
PCB	192.168.0.3	255.255.255.0	192.168.0.1	2001:DB8:1:10::10	FE80::1
R1 G0/0	192.168.0.1	255.255.255.0	-	-	-
R1 G0/1	10.10.10.1	255.255.255.252	-	-	-
Server0	10.10.10.2	255.255.255.252	10.10.10.1	2001:DB8:1:30::30	FE80::30

Teacher's Signature

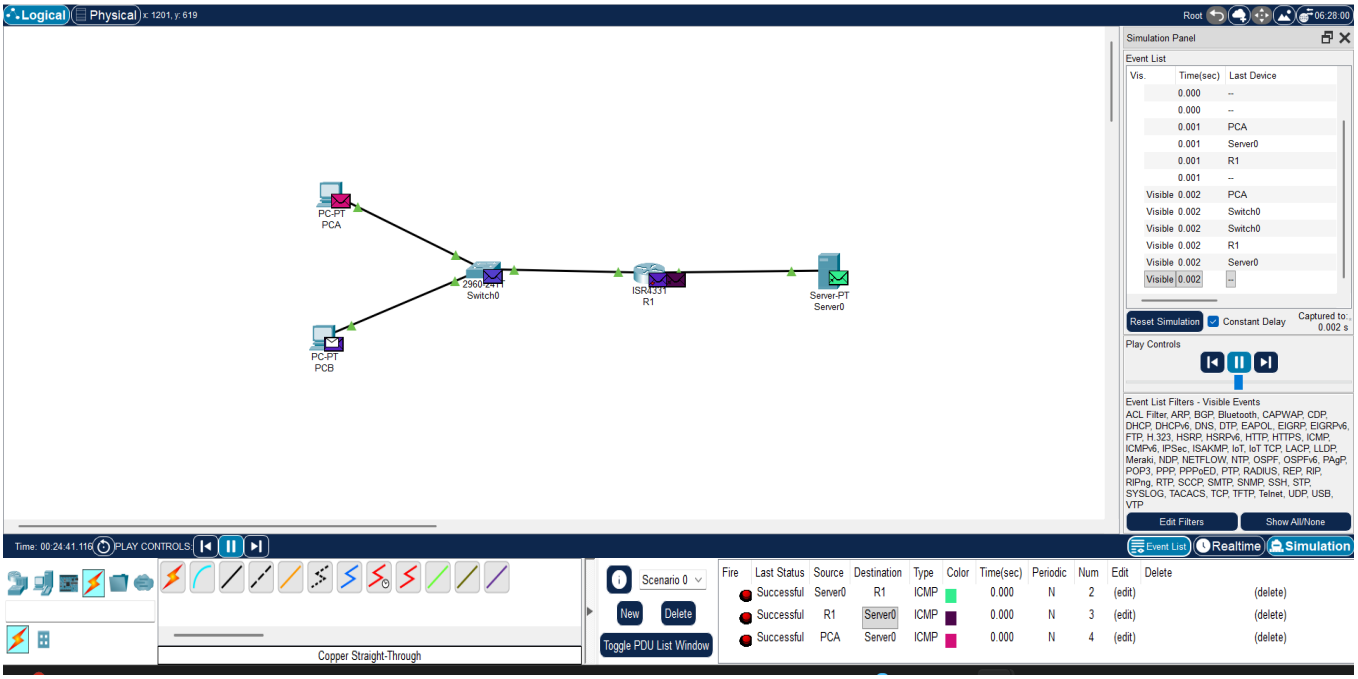
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 07-02-2026**

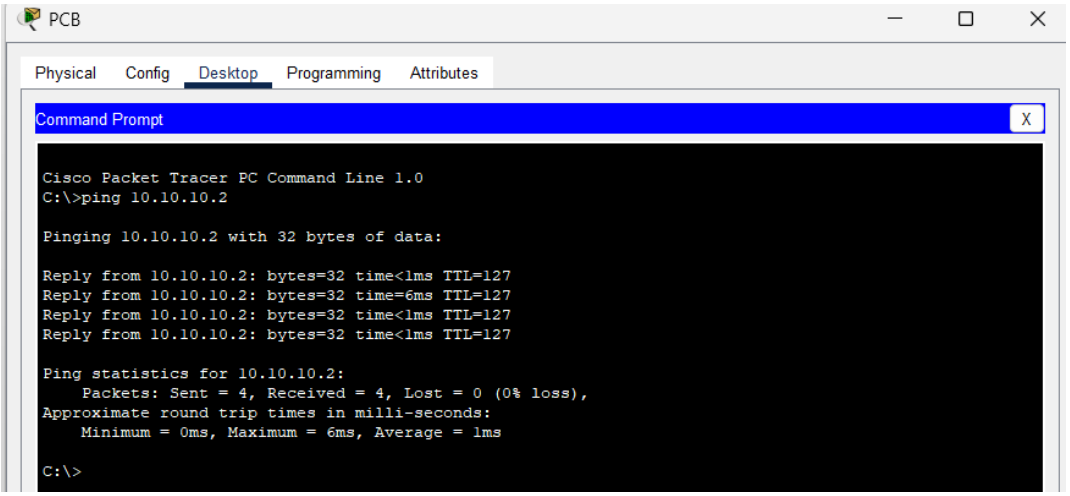
Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 06

5. Now communication is allowed in the network.



6. Click on PCB → Desktop → Command Prompt → write command.
   - ping 10.10.10.2



7. Click on R1 → CLI → paste these lines in that.
   - ip route 192.168.0.0 255.255.255.0 10.10.10.2

```
R1(config-if)#
R1(config-if)#
R1(config-if)#ip route 192.168.0.0 255.255.255.0 10.10.10.2
```

Teacher's Signature

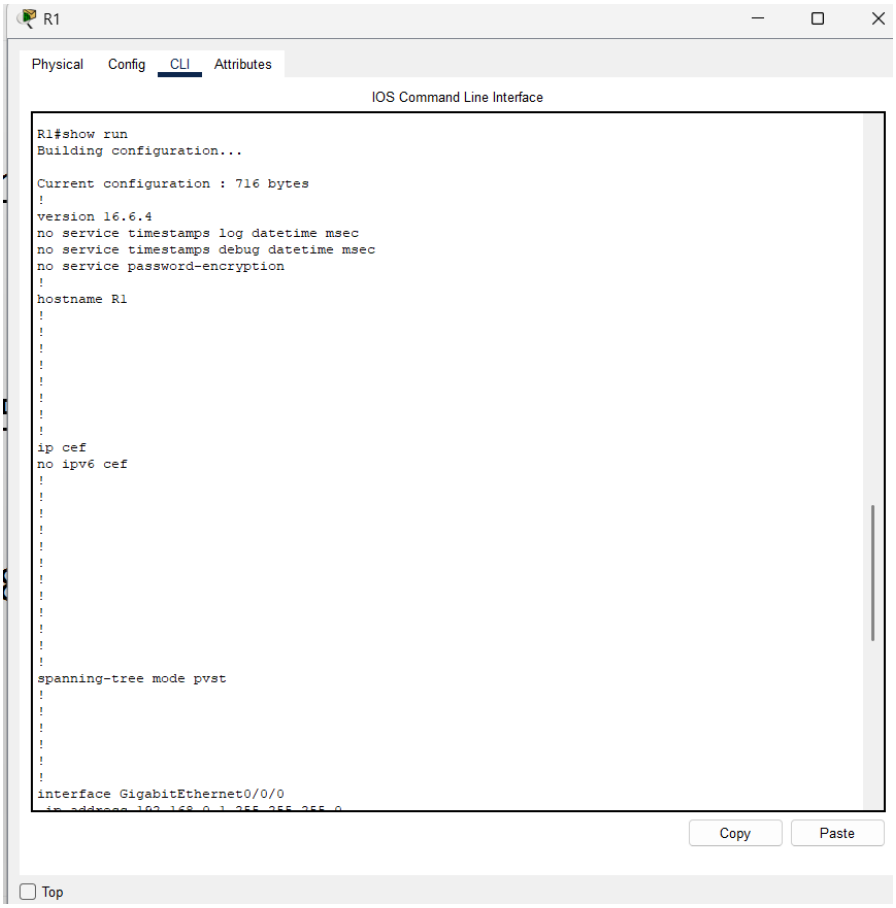
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 07-02-2026**

Course Name: Cybersphere and Security(CSS) **Page no:**

Practical Number:- 06

- show run



- conf t
- enable secret enpa55
- line console 0
- password conpa55
- login
- exit

```
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#enable secret enpa55
R1(config)#line console 0
R1(config-line)#password conpa55
R1(config-line)#login
R1(config-line)#ex
% Ambiguous command: "ex"
R1(config-line)#exit
```

Teacher's Signature

- ip domain-name ccnasecurity.com
- username admin secret adminpa55
- line vty 0 4
- login local
- exit

```
R1(config)#ip domain-name ccnasecurity.com
R1(config)#username admin secret adminpa55
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R1(config)#
```

- crypto key generate rsa
- 1024

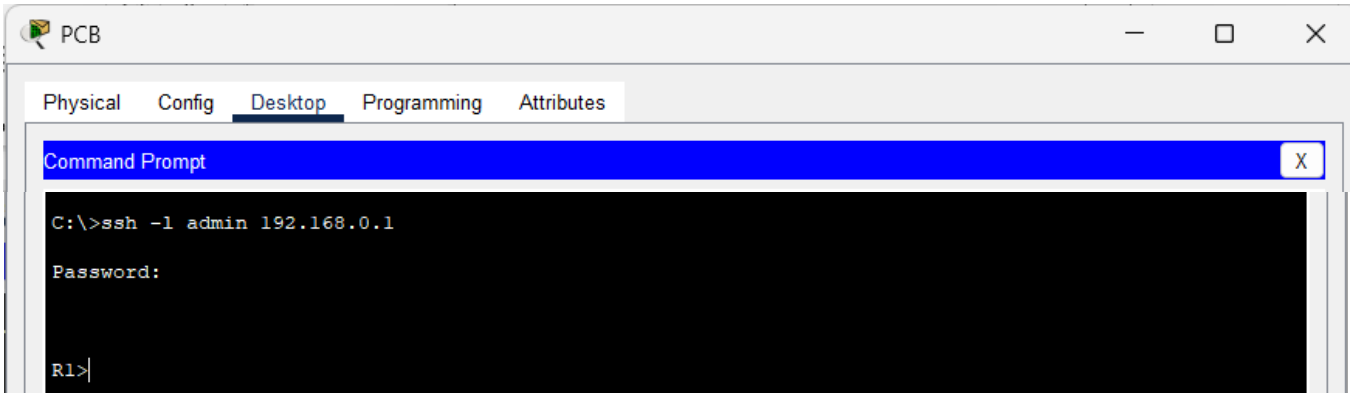
```
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#
```

- access-list 1 permit 192.168.0.2 0.0.0.0
- access-list 1 deny any

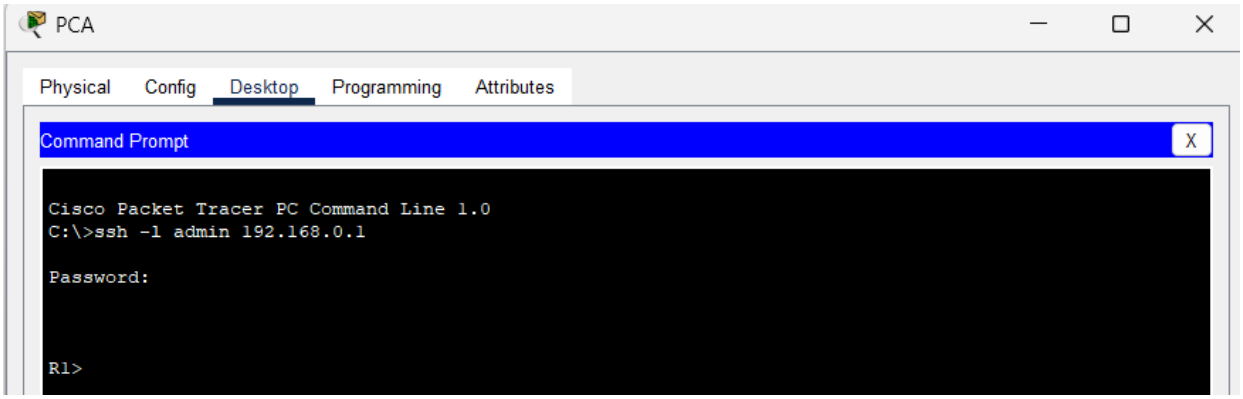
```
R1(config)#access-list 1 permit 192.168.0.2 0.0.0.0
*Mar 1 0:29:14.566: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#access-list 1 deny any
R1(config)#
```

8. Click on PCB → Desktop → Command Prompt → write command.

- ssh -l admin 192.168.0.1
  - When asked for password enter: '<u>adminpa55</u>'



9. Do same for PCA



10. Click on R1 → CLI → paste these lines in that.

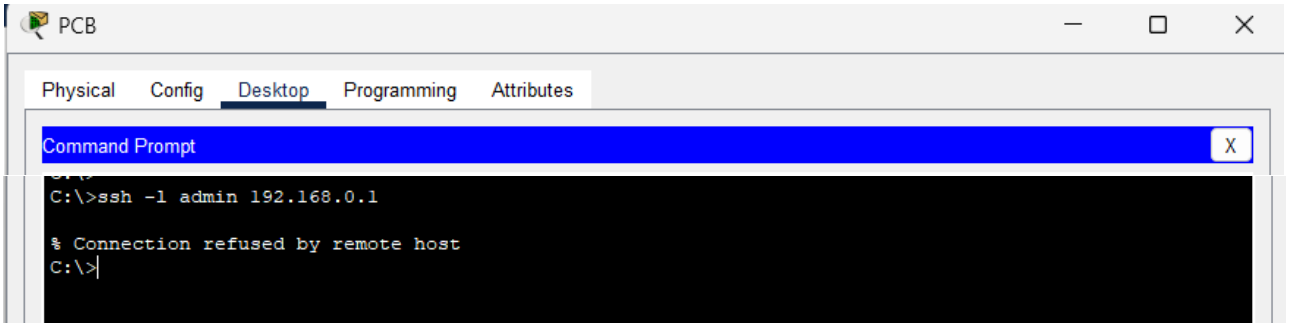
- When asked for password enter: 'conpa55'
- Hit Enter.
  - en
- When asked for password enter: 'enpa55'

- conf t
- line vty 0 4
- access-class 1 in

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 4
R1(config-line)#access-class 1 in
R1(config-line)#
```

11. Click on PCB → Desktop → Command Prompt → write command.

- ssh -l admin 192.168.0.1



12. Click on R1 → CLI → paste these lines in that.

- access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq ftp
- access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq www
- access-list 120 permit tcp host 192.168.0.2 host 10.10.10.2 eq 22
- int gi 0/0/0
- ip access-group 120 in

Teacher's Signature

```
R1(config-line)#access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq ftp
R1(config)#access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq www
R1(config)#access-list 120 permit tcp host 192.168.0.2 host 10.10.10.2 eq 22
R1(config)#int gi 0/0/0
R1(config-if)#ip access-group 120 in
R1(config-if)#
R1(config-if)#
```

13. Click on PCB → Desktop → Command Prompt → write command.

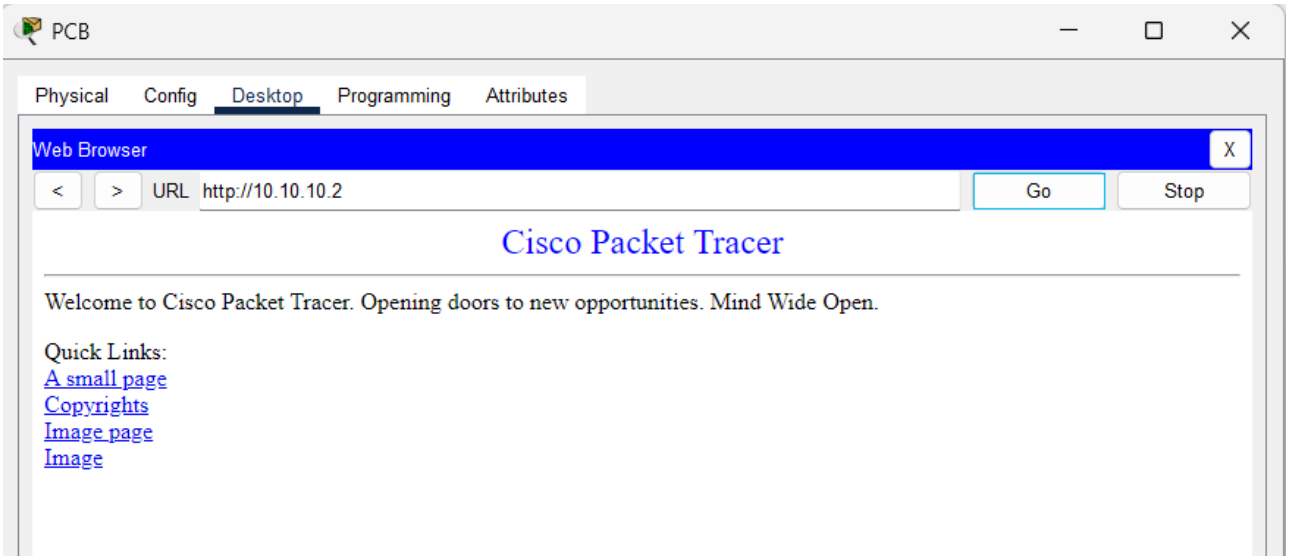
- ftp 10.10.10.2
  - When asked for Username and Password enter: '<u>cisco</u>' for both

```
C:\>ftp 10.10.10.2
Trying to connect...10.10.10.2
Connected to 10.10.10.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

Top

14. Click on PCB → Desktop → Web Browser → write URL.

- http://10.10.10.2



Teacher's Signature